

Advanced Developer Fundamentals

Given a scenario, justify using sharing objects and Apex managed sharing.

Table of Contents

-  [Share Objects](#)
-  [Apex Managed Sharing](#)
-  [Apex Managed Sharing Recalculation](#)
-  [Scenarios and Solutions](#)



After studying this topic, you should be able to:

-  Describe what Apex Managed Sharing is used for in Salesforce.
-  Explain the role of share objects in Apex Managed Sharing.
-  Describe how to set up Apex Managed Sharing for sharing records programmatically.
-  Describe the considerations associated with Apex Managed Sharing and its recalculation.
-  Identify the reasons and use cases associated with the use of Apex Managed Sharing.



Introduction

Apex managed sharing is one of three sharing capabilities in Salesforce. **Managed sharing** access is granted by the system based on record ownership, role hierarchy, and sharing rules. **User managed sharing**, also known as **manual sharing**, allows the record owner or any user with Full Access to a record to share the record with a user or group of users. In situations where neither Managed nor User-managed sharing is suitable, Apex managed sharing can be used. **Apex Managed Sharing** allows you to use **Apex code** to build **sophisticated** and **dynamic** sharing settings that are not otherwise possible.



Apex Managed Sharing



Share Objects

Table of Contents

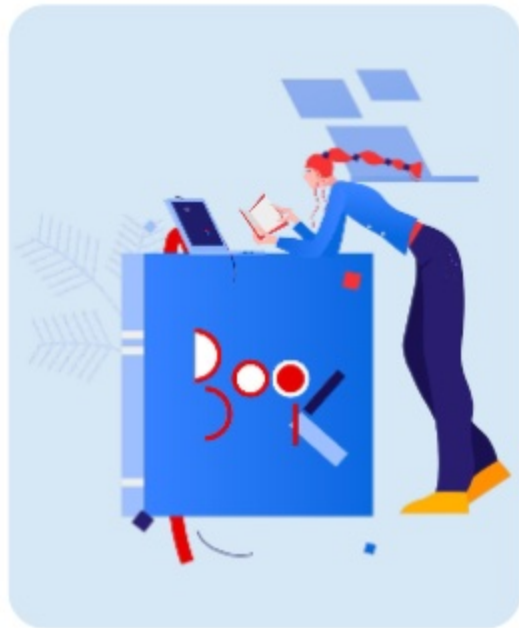


Share Objects

Share objects are used to **create sharing** and **track** records that are being shared in an org.

◆ SHARING SETTING

A Share object **only exists** for an object if the Organization-Wide Default is **Public Read Only** or **Private**. If the object access policy is **Public Read/Write**, then **no other sharing policy** is necessary.



Share Objects

❖ STANDARD SHARES

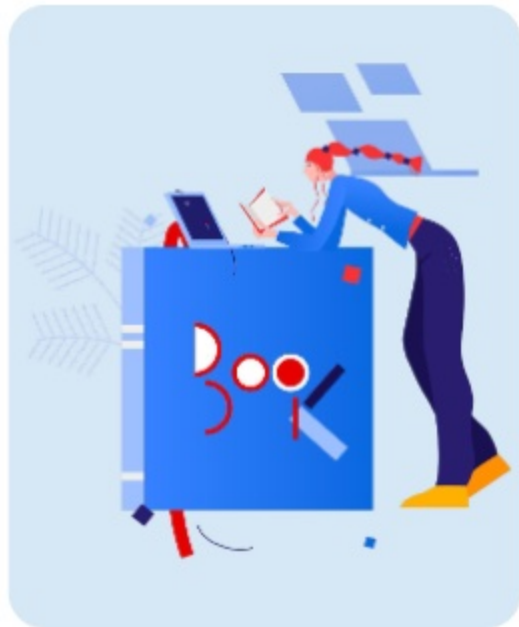
Share objects for **standard objects** append '**Share**' to the **object name**.
For example, the Share object for *Account* would be *AccountShare*.

❖ CUSTOM SHARES

Share objects for **custom objects** append '**_Share**' to the **object name**.
For example, the Share object for *MyCustomObject* would be *MyCustomObject_Share*.

❖ CUSTOM OBJECTS

Although Share objects can exist for both standard and custom object, **Apex sharing reasons** and **Apex managed sharing recalculation** are **only available** for custom objects.



Sharing Objects

Share objects are used in supporting the different sharing types, such as **managed sharing**, **user managed sharing**, and **Apex managed sharing**.



❖ USER MANAGED SHARING

Manually sharing records using Apex are supported by **all custom objects**, but is supported only by a **few standard objects**, such as **Account**, **Contact**, **Opportunity**, and **Case**.

❖ SHARING OBJECT EXISTENCE

A Sharing object **only exists** for an object if the **Organization-Wide Default** is **Public Read Only or Private**. If the object access policy is Public Read/Write, **no other sharing policy** is necessary.

Sharing Objects



◆ APEX MANAGED SHARING

Apex Managed Sharing uses **Apex sharing reason**, which is only supported in **custom objects**, to build sophisticated and **dynamic sharing settings** that are otherwise not possible.

◆ REASONS AND RECALCULATION

Apex sharing reasons and Apex managed sharing recalculation are **only available for custom objects**.

Share Object Definition

When writing the Apex code for sharing a record, the following **Share object fields** for a **custom object** must be defined:



PARENT ID

ParentId corresponds to the ID of the **record** that is being shared to a user or public group.



USER/GROUP ID

UserOrGroupId is the ID of the user or public group who will be **granted access** to the record.



ACCESS LEVEL

AccessLevel, which can be either **'Read'** or **'Edit'**, defines the access level of the user or public group.



ROW CAUSE

RowCause is used for specifying the **reason why** the user or public group is being granted access.

Share Records Example

The following example query returns a list of **share records** for the standard **Account** object.

```
SELECT AccountId, RowCause, AccountAccessLevel, ContactAccessLevel, CaseAccessLevel, OpportunityAccessLevel, UserOrGroupId FROM AccountShare
```

Query Results - Total Rows: 127

AccountId	RowCause	AccountAccessLevel	ContactAccessLevel	CaseAccessLevel	OpportunityAccessLevel	UserOrGroupId
0012v00002RVfY3AAL	Owner	All	Edit	Edit	Edit	0052v00000Z2QwQAAV
0012v00002RVfYsAAL	Owner	All	Edit	Edit	Edit	0052v00000Z2QwQAAV
0012v00002RVfYtAAL	Owner	All	Edit	Edit	Edit	0052v00000Z2QwQAAV
0012v00002MZorbAAD	Owner	All	Edit	Edit	Edit	0052v00000Z2QwRAAV
0012v00002MZorgAAD	Owner	All	Edit	Edit	Edit	0052v00000Z2QwTAAV
0012v00002MZoreAAD	Owner	All	Edit	Edit	Edit	0052v00000Z2QwUAAV
0012v00002MZordAAD	Owner	All	Edit	Edit	Edit	0052v00000Z2QwVAAV
0012v00002MZorfAAD	Owner	All	Edit	Edit	Edit	0052v00000Z2QwVAAV
0012v00002RUViPAAX	Manual	Read	Read	Read	Read	00G2v000003gYOoEAM
0012v00002RUViPAAX	Manual	Edit	None	None	None	00G2v000004W2Q6EAK

Sharing Records



NOTES



Salesforce allows for skipping Implicit Share records for Cases, Contacts, and Opportunities associated with Accounts, resulting in faster ownership and sharing rule calculations for account-related data.

Apex Managed Sharing

Table of Contents



Apex Managed Sharing Overview

Apex Managed Sharing allows developers to build **sophisticated** and **dynamic** sharing settings.



SHARING REASON

Apex Managed Sharing is a type of programmatic sharing that **allows** defining a **custom sharing reason** for the **programmatic share**.



CUSTOM OBJECT

Custom sharing reasons **can only be defined** for **shares** written to **custom objects**.



SHARING SETTING

Objects with a default sharing setting of **'Private'** or **'Public Read Only'** have a related **'Share'** object.



SHARE OBJECTS

A share object includes records supporting **managed sharing**, **user managed sharing**, and **Apex managed sharing**.

User Managed Sharing Using Apex

User managed sharing, or **manual sharing**, can also be performed **programmatically**.



MANUAL SHARING

Apex can be used to perform manual sharing of records of **custom objects** and a **few standard objects** such as **Account**, **Contact**, **Opportunity**, and **Case**.



OWNER CHANGES

The **RowCause** defined is **"Manual"**, and shares using this type of sharing are **automatically deleted** when the record owner is changed.

Using Apex Managed Sharing

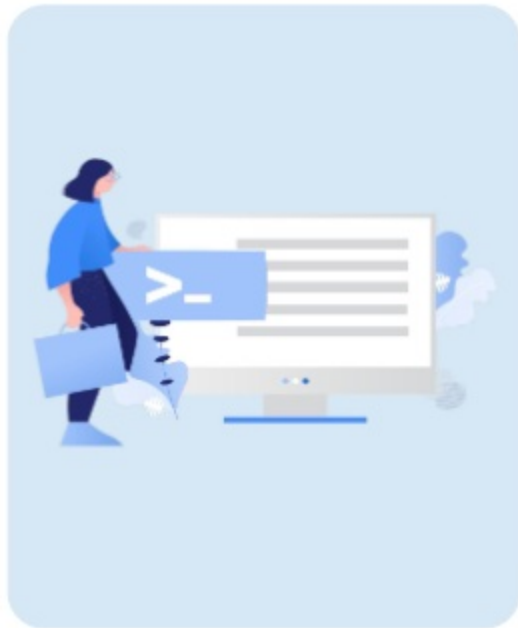
The following describes details for **implementing** for Apex Managed Sharing:

❖ APEX SHARING REASON

To use Apex managed sharing, a **sharing reason must first be defined** for the custom object. Apex code can then be written to add or remove entries of the Share object to **grant users** access to a record when specific conditions are met.

❖ OBJECT DEFINITION

Apex sharing reasons can be **created** by navigating to the '**Apex Sharing Reasons**' related list of the custom object in the **Salesforce Classic** interface.



Using Apex Managed Sharing

◆ LABEL AND NAME

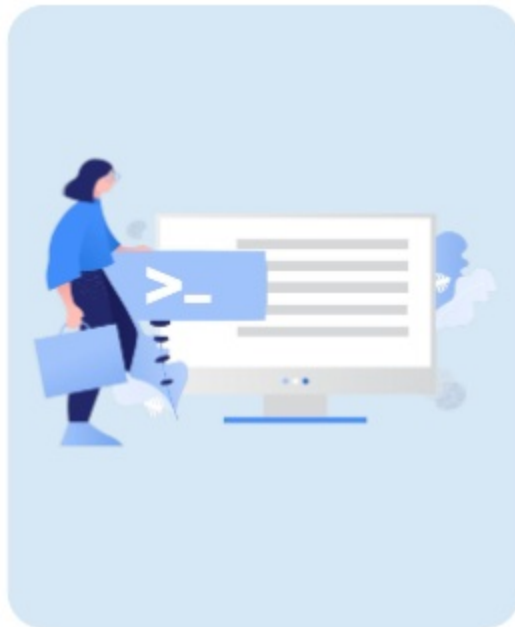
Each Apex sharing reason has a **label** and a **name**. Label is displayed in the **user interface** while viewing the sharing for a record, and the name is used when referencing the reason in the **API** and **Apex**.

◆ ROWCAUSE TYPE

Either a **custom** or '**Manual**' RowCause can be used in Apex code to create records in share tables. In Apex managed sharing, a **sharing reason** must be used.

◆ MANUAL ROWCAUSE

A '**Manual**' RowCause is used to treat the share record as if it were created by **manual sharing**. This value is used in declarative or programmatic **user managed sharing**.



Apex Managed Sharing

Apex Sharing Reasons and **Apex Sharing Recalculation** can be managed in the detail page of a custom object in the **Salesforce Classic** interface.

Record Types

NewPage Layout AssignmentRecord Types Help ?

No record types defined

Apex Sharing Reasons

NewApex Sharing Reasons Help ?

Action	Label	Modified By
Edit Del	Share with Specialist	Martin Gessner, 9/22/2020 5:17 PM

Apex Sharing Recalculation

NewRecalculate Apex SharingApex Sharing Recalculation Help ?

No Apex sharing recalculations defined

Object Limits

Object Limits Help ?

Item	Usage	Limit	% used	Message
------	-------	-------	--------	---------

Apex Managed Sharing Example

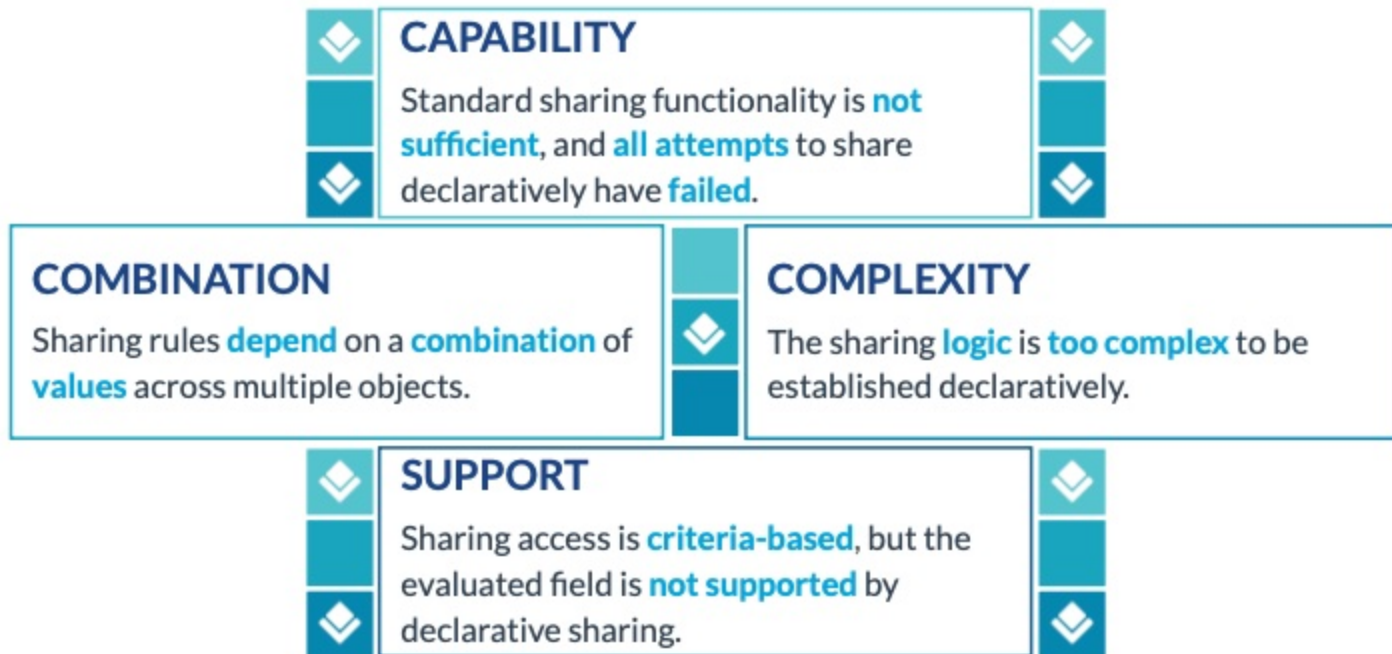
The following Apex function example uses **Apex managed sharing** to share a record by specifying a **record ID** and ID of a **User** or **Group**.

```
public static void manualShareReadWrite(Id recordToShareId, Id shareWithId){  
  
    // Create new sharing object  
    MyCustomObject__Share shareObject = new MyCustomObject__Share();  
  
    //ID of the object we want to share  
    shareObject.ParentId = recordToShareId;  
  
    //ID of the person we want to share with  
    shareObject.UserOrGroupId = shareWithId;  
  
    // Access level - valid values are Read, Edit or ALL.  
    // Note that unlike Read and Edit, ALL is an internal value and cannot be granted.  
    shareObject.AccessLevel = 'Edit';  
  
    // Fill in this field with an existing, manually created sharing reason  
    // defined in the 'Apex Sharing Reasons' section of the custom object definition  
    shareObject.RowCause = Schema.MyCustomObject__Share.RowCause.Share_with_Specialist__c;  
  
    // Activate the defined access to the record  
    insert shareObject;  
}
```

An Apex Sharing Reason is used for tracking or determining why a record was shared with a user or group of users.

When to Use Apex Managed Sharing

The **following conditions** require the use of Apex managed sharing:



Apex Managed Sharing Recalculation

Table of Contents



Apex Managed Sharing Recalculation

Apex managed sharing rules are **automatically recalculated** whenever a custom object's organization-wide sharing default **access level** is **updated**.

◆ CLASSES EXECUTION

During automatic recalculation, **custom recalculation classes** are **executed**.

◆ MANUAL INVOCATION

Sharing rule **recalculation** can be **manually invoked** from the custom object's setup area.



Apex Managed Sharing Recalculation

❖ RECALCULATION CLASS

To **enable manual recalculation**, an Apex class that implements the **Database.Batchable interface** must be written.

❖ OBJECT ASSOCIATION

To associate an Apex managed sharing **recalculation class** with a **custom object**, the recalculation class is selected in the **Apex Sharing Recalculations** section of the object definition.



Apex Managed Sharing Considerations

The following are some key **considerations** when **implementing** Apex managed sharing:



◆ MOST PERMISSIVE

When **multiple entries** of a Share object apply to a certain user, the **most permissive** rule is applied. For example, if a user has been granted *Read* access to the record through **managed** or **manual sharing**, and an **Apex managed sharing** rule grants *Read/Write* access, then *Read/Write* access is applied.

Apex Managed Sharing Considerations



❖ OWNER CHANGES

Apex managed sharing is **maintained** across **record owner changes**.

❖ MODIFY ALL DATA

Only users with the 'Modify All Data' permission can **add**, **edit**, or **delete** Apex managed sharing on a record.

❖ SHARING REASONS

A record can be **shared multiple times** with a user or group using **different** Apex sharing reasons.

Scenarios and Solutions

Table of Contents



Apex Managed Sharing Use Cases



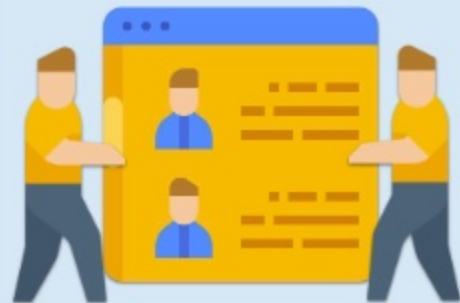
SCENARIO

Once a Job record is created by a Recruiter, it needs to be shared automatically with a hiring manager based on the value in a lookup field on the Job record.



SOLUTION

In this case, an **'after insert'** trigger or a **record-triggered flow** can be created to share a newly created Job record with its associated hiring manager automatically. A **Share record** for the Job record would need to be created with *ParentId*, *UserOrGroupId*, *AccessLevel*, and *RowCause*. The *RowCause* can be specified by navigating to the **'Apex Sharing Reasons'** related list of the Job custom object.



Apex Managed Sharing Use Cases



SCENARIO

Records of a custom object called *Expense* in an organization need to be shared to users based on values of certain custom fields on the Contact records which they own.



SOLUTION

Apex Managed Sharing allows using Apex to share **custom object** records programmatically. If the sharing logic is too complex and/or involves other objects, Apex code can be defined to create and insert **share records** for the custom object records. In this scenario, the **share object** for Expense would be *Expense_Share*.

Apex Managed Sharing Use Cases



SCENARIO

An organization needs to grant access to users to the Warehouse custom object based on different criteria. The reason for granting access needs to be visible in the user interface.

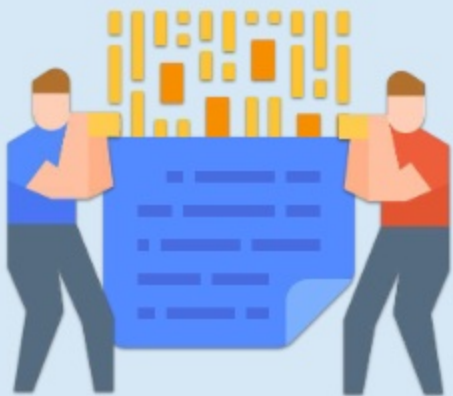


SOLUTION

Apex Sharing Reasons can be defined for a custom object from the 'Apex Sharing Reasons' related list. It can be used while creating Apex code to provide access to **custom object** records programmatically. A reason can be specified while assigning the value of the *RowCause* of the share record. The reason can be accessed using `Schema.CustomObjectName_Share.RowCause.SharingReason_c`.



Apex Managed Sharing Use Cases



SCENARIO

When a lead is converted into an opportunity, the opportunity needs to be shared with the lead owner, if the record meets certain criteria based on computed values from certain fields.



SOLUTION

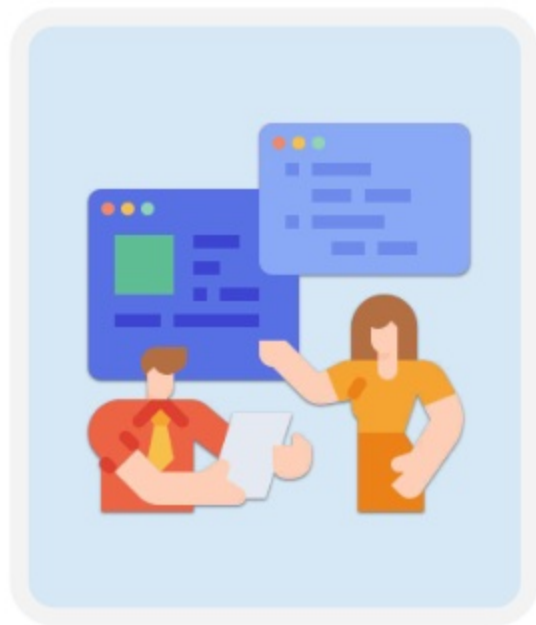
The **Opportunity** records created from the leads can be shared **programmatically** with the lead owner using Apex code. However, the **RowCause** would need to be **'Manual'**, since custom sharing reasons are only supported by custom objects. This scenario illustrates a programmatic implementation of **manual sharing** for a record.

Apex Managed Sharing Use Cases



SCENARIO

The organization-wide default sharing setting of a custom object named 'Promotion__c' has been set to 'Private'. A developer has created an Apex trigger that shares a record of the custom object programmatically with a marketing manager if the record meets a certain set of criteria.



Apex Managed Sharing Use Cases

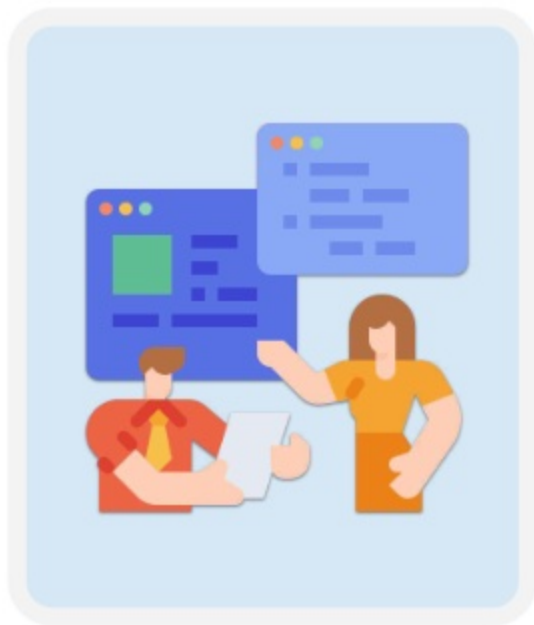


SOLUTION

When the organization-wide default sharing setting of a custom object such as 'Promotion__c' is set to Private, a '**share object**' exists for the object.

In this scenario, the organization will have a share object named 'Promotion__Share', which can be utilized to share a record of the corresponding custom object using **Apex Managed Sharing**.

Sharing a record of the object with the marketing manager using an Apex trigger will result in the existence of a share record in the 'Promotion__Share' table. The **RowCause** of the share record will be defined with an **Apex sharing reason** when creating the share record.



Learn More

-  [Sharing a Record Using Apex](#)
-  [Just-In-Time Sharing Slide Deck from DeveloperForce](#)
-  [Understanding Sharing](#)
-  [Recalculating Apex Managed Sharing](#)

